Phys 115B HW 6

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February 13, 2026

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Question 1. *1 Normalization (Griffiths 5.4, same)*

(a) If ψ_a and ψ_b are orthogonal, and both are normalized, what is the constant A in equation $\psi_\pm(\mathbf{r}_1, \mathbf{r}_2) = A(\psi_a(\mathbf{r}_1)\psi_b(\mathbf{r}_2) \pm \psi_b(\mathbf{r}_1)\psi_a(\mathbf{r}_2))$?

(b) If $\psi_a = \psi_b$ (and it is normalized), what is A? (This case, of course, occurs only for bosons.)

Proof.

(a) Consider the normalization condition of individual states, we get the following equation:

$$\forall u, v \in \{a, b\}, \quad \langle \psi_u | \psi_v \rangle = \delta_{uv} \tag{1}$$

Which, if consider the states as tensor product, then one has the following (we'll use $|\psi\rangle_1$ to indicate $\mathbf{r}_1$, and $|\psi\rangle_2$ to indicate $\mathbf{r}_2$):

$$|\psi_\pm\rangle = A(|\psi_a\rangle_1 \otimes |\psi_b\rangle_2 \pm |\psi_b\rangle_1 \otimes |\psi_a\rangle_2) \tag{2}$$

So, one gets the following regarding the normalization condition:

$$\begin{aligned} 1 &= \langle \psi_\pm | \psi_\pm \rangle && (3)\\ &= |A|^2(\langle \psi_a|\psi_a\rangle_1 \cdot \langle \psi_b|\psi_b\rangle_2 + \langle \psi_b|\psi_b\rangle_1 \cdot \langle \psi_a|\psi_a\rangle_2 \pm \langle \psi_b|\psi_a\rangle_1 \cdot \langle \psi_a|\psi_b\rangle_2 \pm \langle \psi_a|\psi_b\rangle_1 \cdot \langle \psi_b|\psi_a\rangle_2) && (4)\\ &= |A|^2(1\cdot 1 + 1 \cdot 1 \pm 0 \cdot 0 \pm 0 \cdot 0) = 2|A|^2 && (5) \end{aligned}$$

So, one gets that $|A| = \frac{1}{\sqrt{2}}$. Take the positive real value, take $A = \frac{1}{\sqrt{2}}$ is the normalization constant.

(b) If $\psi = \psi_a = \psi_b$, then the state $\psi_+ = 2A\psi(\mathbf{r}_1)\psi(\mathbf{r}_2)$. Which, consider the normalization, one has the following:

$$1 = \langle \psi_+ | \psi_+ \rangle = 4|A|^2 \langle \psi|\psi\rangle_1 \cdot \langle \psi|\psi\rangle_2 = 4|A|^2 \tag{6}$$

So, one gets $|A|^2 = \frac{1}{4}$, or $|A| = \frac{1}{2}$. Take A as positive real, one has $A = \frac{1}{2}$.

□

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Question 2. *2 Particles in the infinite square well (Griffiths 5.5, same)*

(a) *Write down the Hamiltonian for two noninteracting identical particles in the infinite square well. Verify that the fermion ground state given by* $\frac{\sqrt{2}}{a}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right)$ *is an eigenfunction of* $\hat{H}$*, with the appropriate eigenvalue.*

(b) *Find the next two excited states (beyond the ones given in the example)–wave functions, energies, and degeneracies–for each of the three cases (distinguishable, identical bosons, identical fermions).*

Hamiltonian (when restricting $x_1, x_2 \in [0, a]$*):*

$$\hat{H} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}\right)$$

Proof.

(a) Applying $\hat{H}$ given above to the given fermion ground state, we get:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}\right)\left(\frac{\sqrt{2}}{a}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right)\right) \tag{7}$$

$$= -\frac{\hbar^2}{\sqrt{2}am}\left(-\frac{\pi^2}{a^2}\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) + \frac{4\pi^2}{a^2}\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right) \tag{8}$$

$$-\frac{\hbar^2}{\sqrt{2}am}\left(-\frac{4\pi^2}{a^2}\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) + \frac{\pi^2}{a^2}\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right) \tag{9}$$

$$= \frac{5\hbar^2\pi^2}{\sqrt{2}a^3m}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right) \tag{10}$$

$$= \frac{5\hbar^2\pi^2}{2ma^2}\cdot\left(\frac{\sqrt{2}}{a}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right)\right) \tag{11}$$

This verifies that the given function is indeed an eigenfunction of $\hat{H}$, with eigenvalue $\frac{5\hbar^2\pi^2}{2ma^2}$.

(b) Here are the cases for each case:

– **Distinguishable:**

Since the distinguishable states are $\psi_{n_1 n_2} = \psi_{n_1}(x_1)\psi_{n_2}(x_2)$ with eigenvalue $E_{n_1 n_2} = (n_1^2 + n_2^2)K$ (for $K = \frac{\hbar^2\pi^2}{2ma^2}$). With the case of $(n_1, n_2) = (1,1), (1,2), (2,1)$ being described (corresponds to $n_1^2 + n_2^2 = 2, 5, 5$ respectively), the next two eigenvalues should be $(n_1, n_2) = (2,2)$ (corresponding to $E_{22} = 8K$), and $(n_1, n_2) = (1,3), (3,1)$ (corresponding to $E_{13} = E_{31} = 10K$)

□

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Question 3. *3 Exchange forces across orbitals (Griffiths 5.6, same)*

Proof. □

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Question 4. *4 Three particles (Griffiths 5.8, 5.7 in second edition)*

Proof. □

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Question 5. *5 Spins in the SHO (similar to Griffiths 5.9 in third edition)*
Consider two non-interacting, identical particles of mass m moving in a one-dimensional simple harmonic oscillator potential of frequency ω. Determine the energies and wavefunctions of the ground state and the first excited state for
(a) two spin 1/2 fermions
(b) two spin 1 bosons

Proof. □

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Question 6. *6 Atoms in a box*
Suppose we have a 3D cubic box with sides of length L. The potential is zero inside the box and infinite outside. We place some number N of indistinguishable non-interacting particles of mass m and spin S into the box.
As a reminder, the allowed energies for a single particle in the 1D box are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \equiv n^2 E_1$$

(a) What is the ground state degeneracy for $N = 3$, and $S = 0$?
(b) What is the ground state degeneracy for $N = 3$, and $S = 1/2$?
(c) What is the ground state degeneracy for $N = 2$, and $S = 1$?
(d) What is the ground state degeneracy for $N = 4$, and $S = 3/2$?

Question 7. *7 The Ground State of Lithium (Griffiths 5.16, DNE in second edition)*
Ignoring electron-electron repulsion, construct the ground state of lithium ($Z = 3$). Start with a spatial wave function, remembering that only two electrons can occupy the hydrogenic ground state; the third goes to $\psi_{2,0,0}$. What is the energy of this state? Now tack on the spin, and antisymmetrize. What's the degeneracy of the ground state?
Hint: Griffiths 5.10 (doesn't exist in second edition) might be very useful.

Proof. □